AWS Secure 3-Tier Architecture Documentation

OpenTofu >= 1.6.0 AWS: ap-southeast-5 WAFv2: Enabled Graviton: ARM64

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Welcome to the official technical documentation for our **AWS 3-Tier Deployment for AI & Web Infra** project. This project is optimized for deployment in the **AWS Asia Pacific (Malaysia) region (ap-southeast-5)** utilizing AWS Graviton (ARM64) instances, Multi-AZ RDS Postgres, and AWS WAFv2 regional protection.

This deployment is structured natively in OpenTofu, adhering to strict modular boundaries, security best practices, and the **Zero-Trust Network Principle**.

Technical Overview

The architecture divides infrastructure components into discrete logical and physical tiers to achieve top-tier scalability, performance, and threat mitigation:

- **Presentation / Web Layer:** Application Load Balancer (ALB) receiving external traffic and filtered through AWS WAFv2 (OWASP rules + rate limiting).
- **Application / Compute Layer:** Auto Scaling Group (ASG) of EC2 instances running inside private subnets, auto-scaled on CPU usage, and managed via AWS Systems Manager (SSM).
- **Database Layer:** Multi-AZ RDS PostgreSQL database deployed within isolated subnets, allowing connections exclusively from the application tier.

Documentation Index

Explore different sections of our infrastructure documentation:

Core Configuration

- **System Architecture**: Deep dive into the physical network structure, routing tables, and AWS resource layout in the Malaysia region, including how the Developer’s first design is mapped.
- **AWS Phased Adoption Roadmap & Costing Guide**: Multi-year week-by-week and month-by-month AWS service growth plan mapped from the project Gantt chart.
- **Developer Design Alignment Guide**: Rationale and comparison of shifting from legacy single-node Ubuntu VMs to an enterprise secure AWS design.
- **ASGs & Separation of Concerns Guide**: Best practice guide detailing auto-scaling with distinct ASGs, stateless principles, and the role of Amazon S3, EFS, or both.
- **Root OpenTofu/Terraform Files**: Overview of the root configuration entries (`main.tf` , `variables.tf` , `outputs.tf` , `providers.tf`).
- **OpenTofu Migration Guide**: Detailed compatibility research and transition path for deploying using OpenTofu on AWS.
- **AMI Design & Hardening Guide**: Architectural guide outlining the pre-baked AMI strategy, Packer/Ansible pipeline, and ASIMP compliance integration.
- **Route 53 & DNS Troubleshooting**: Deep dive into custom domain integration, ACM SSL/TLS setup, and extensive research on resolving ASG private subnet DNS resolution failures.
- **Secure Developer Access Guide**: Comprehensive guide on using our secure SSH Jumphost (Bastion) to access private and standalone nodes from Cyberjaya, with Windows/macOS/Linux client setups and private key security guidelines.
- **Hybrid Cloud Integration Guide**: Comprehensive evaluation of secure, cost-optimized API and MCP connections alongside official

AWS hybrid network solutions (VPN, Direct Connect, Transit Gateway) with granular MYR/USD costing models for ap-southeast-5.

- **Disaster Recovery Options & National Sovereignty Guide**: Production-ready playbook covering the 4 standard AWS cloud DR options aligned with our Multi-AZ architecture, detailed local sovereignty/PDPA/CBPDT compliance reviews, and granular USD/MYR costing comparisons.
- **RDS PostgreSQL 17 vs. Percona Server for PostgreSQL 17 Guide**: Comprehensive technical comparison of performance, telemetry, observability (PMM vs CloudWatch/Performance Insights), architectural designs, and costs in USD/MYR for ap-southeast-5.
- **RAGFlow + Langfuse GPU Guide**: Architectural and economic analysis of RAGFlow and Langfuse, detailing critical GPU utilization, AWS vs. On-Premises deployment models, and secure API/MCP hybrid connections.
- **Google Antigravity Skills Guide**: Comprehensive integration guide outlining how to share and deploy agent knowledge bases and skills between Google Jules and Google Antigravity.
- **SOP: Knowledge-First Discovery**: Standard Operating Procedure outlining how AI agents perform local documentation search before probing remote targets.
- **Wazuh Standalone Cloud Installation & Costing Guide**: Detailed installation steps for standalone Wazuh in the cloud (AWS Marketplace AMI and Graviton assistant modes), with independent USD/MYR costing tables for isolated financial tracking.
- **Technology Stack Comparison Guide**: Complete architectural mapping and AWS alternatives comparison for the developer’s containerized and external integrations (LangChain4j, Spring Boot, React, Twilio, Meta, Postgres, Valkey).
- **Software Licensing & Technology Risk Register (TS/MC Series)**: Complete software licensing compliance framework, technology risk registry, and mitigation plans (TS/MC Series) covering LangChain4j, self-hosted operations, Bedrock with Qwen3 models, and standalone Wazuh SIEM.
- **Strategic Comparative Review**: Comprehensive strategic analysis and financial TCO comparison of an AWS-Native Managed Platform

against a Self-Hosted / On-Premises Custom Stack in Malaysia.

- **Load Testing Assumptions & Sizing Guide**: Workload definitions, SLA metrics, architectural performance assumptions, and multi-VU sizing models from 100 to 10,000 VUs.

Infrastructure Submodules

- **VPC Module**: Core networking, public/private subnets, internet gateways, and NAT configurations.
- **Security Groups Module**: Strict firewall rulesets and port-level isolation.
- **WAF Module**: Layer-7 Web Application Firewall protecting the ALB.
- **ALB Module**: Application Load Balancer and health-check configurations.
- **ASG Module**: Auto Scaling Group, Launch Templates, and dynamic Graviton auto-detection.
- **RDS Module**: Multi-AZ PostgreSQL configuration and parameter group tuning.
- **Standalone EC2 Module**: Secure standalone Ubuntu 26.04 LTS development and application environments.
- **ElastiCache Valkey Module**: Secure ElastiCache Valkey in-memory caching cluster for session and metadata store.
- **Jumphost Module**: Secure public-subnet SSH Jumphost (Bastion) whitelisted for Cyberjaya office with automated downstream ingress configuration.

Deployment & CI/CD

- **Automation Scripts**: Details about CLI helpers (`deploy.sh`, `destroy.sh`, `user_data.sh`).
- **CI/CD Pipeline**: GitHub Actions workflow for automatic formatting, testing, validation, and OIDC deployment.
- **GitLab EFS CI/CD**: Comprehensive guide on GitLab CI/CD, automatic workflows, EFS mounting, dynamic Nginx path configurations, and containerized/S3 alternatives.

- **Costing Estimate**: Comprehensive monthly cost breakdown, local currency estimates, and Day-2 cost optimization pathways.

Prerequisites

Before deploying the infrastructure, ensure you have the following tools installed and configured:

- **OpenTofu >= 1.6.0** (Recommended) or **Terraform >= 1.5.0**
- **AWS CLI** configured with admin-level credentials for `ap-southeast-5`
- **Git** for repository and revision tracking